



Présentation Projet SMACP

Mathias DIJOUX

Willan BANOR

Eliott MOLARD

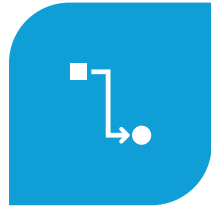
Sommaire



CONTEXTE /
PROBLÉMATIQUE



OBJECTIFS DU
PROJET



DIAGRAMMES



ARCHITECTURE
GÉNÉRALE



TECHNOLOGIE
UTILISÉES



FONCTIONNEMENT
DE L'APPLICATION



INTERFACE
UTILISATEUR



DIFFICULTÉS
RENCONTRÉES



CONCLUSION

Contexte / Problématique

Etude du
Laboratoire
PIMENT Université
de la Réunion

Fonctionnel de
2019 - 2024

Implémentation
par le SIDELEC
dans le cirque de
Mafate

Répartition des tâches

Étudiant 1 : Mathias DIJOUX

- Acquisition des données électriques des panneaux photovoltaïque
- Acquisition des données de rayonnement solaire et données environnementales in situ
- Déploiement de la solution Arduino Wifi
- Détection de défauts et mise en protection
- Cybersécurisation et tableau de bord de la partie production électrique

Étudiant 2 : Willan BANOR

- Acquisition des données du pack de batterie et du contrôle de la charge
- Cartographie GPS pour la position de la centrale
- Déploiement de la solution type bus de terrain (CAN ou RS485)
- Détection des défauts et mise en protection
- Cybersécurisation et tableau de bord de la partie stockage

Étudiant 3 : Eliott MOLARD

- Tableau de bord administrateur
- Infrastructure réseau LAN / IoT
- Intégration des solutions E1 ,E2
- Cybersécurisation de la solution

Objectif du projet

- Besoin de supervision à distance
- Centraliser les données
- Sécuriser l'installation
- Surveiller les performances

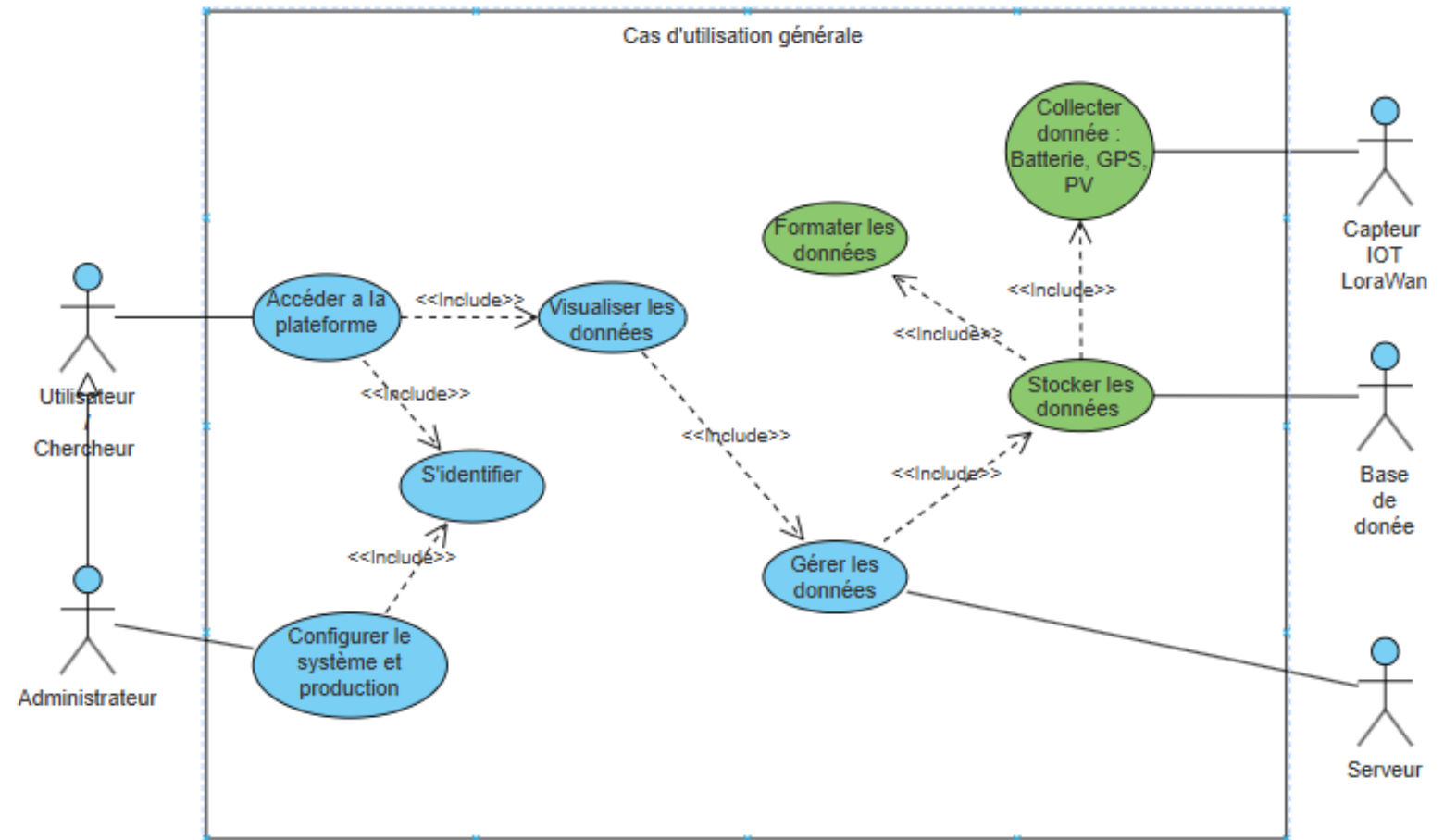
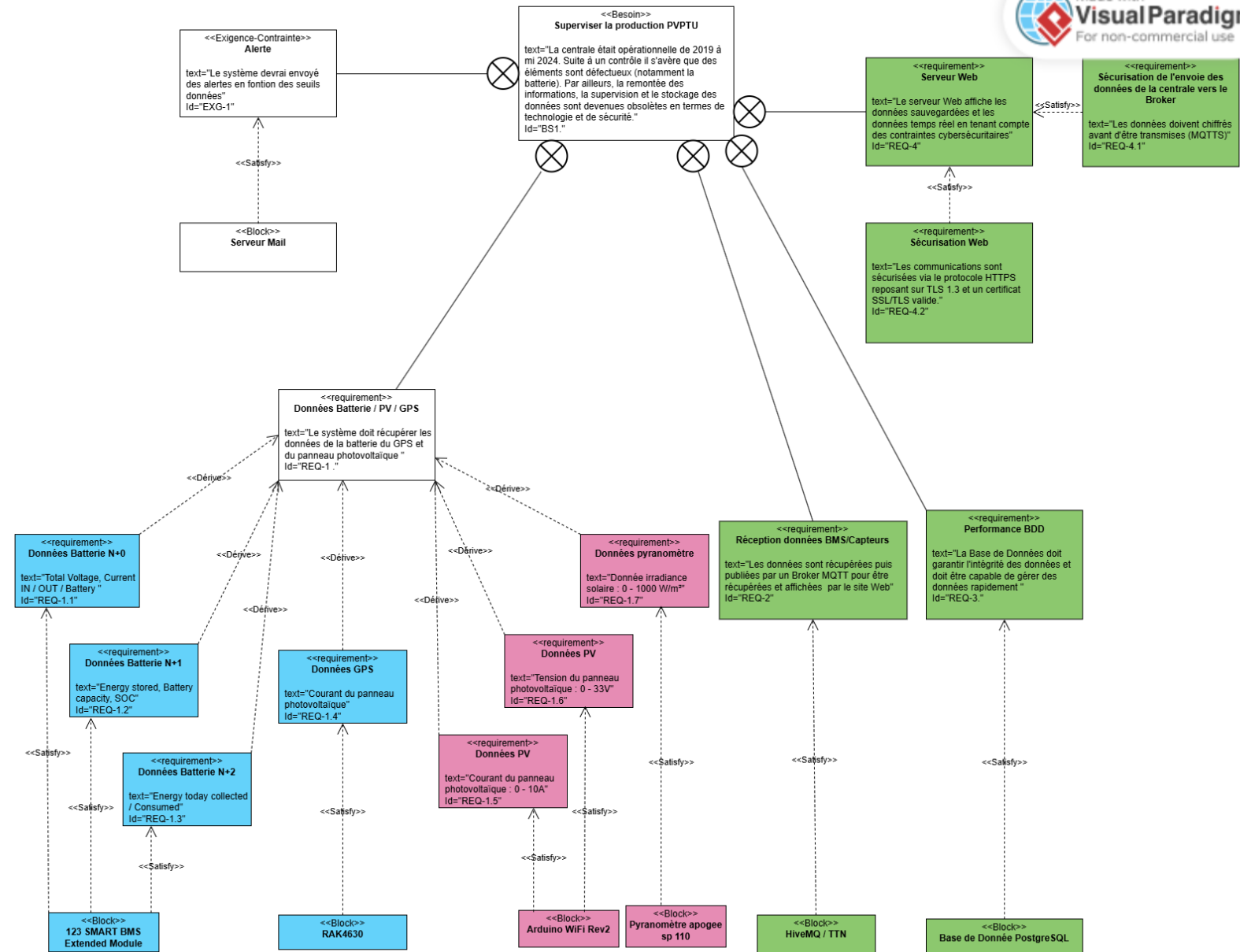


Diagramme d'exigence



Synoptique projet

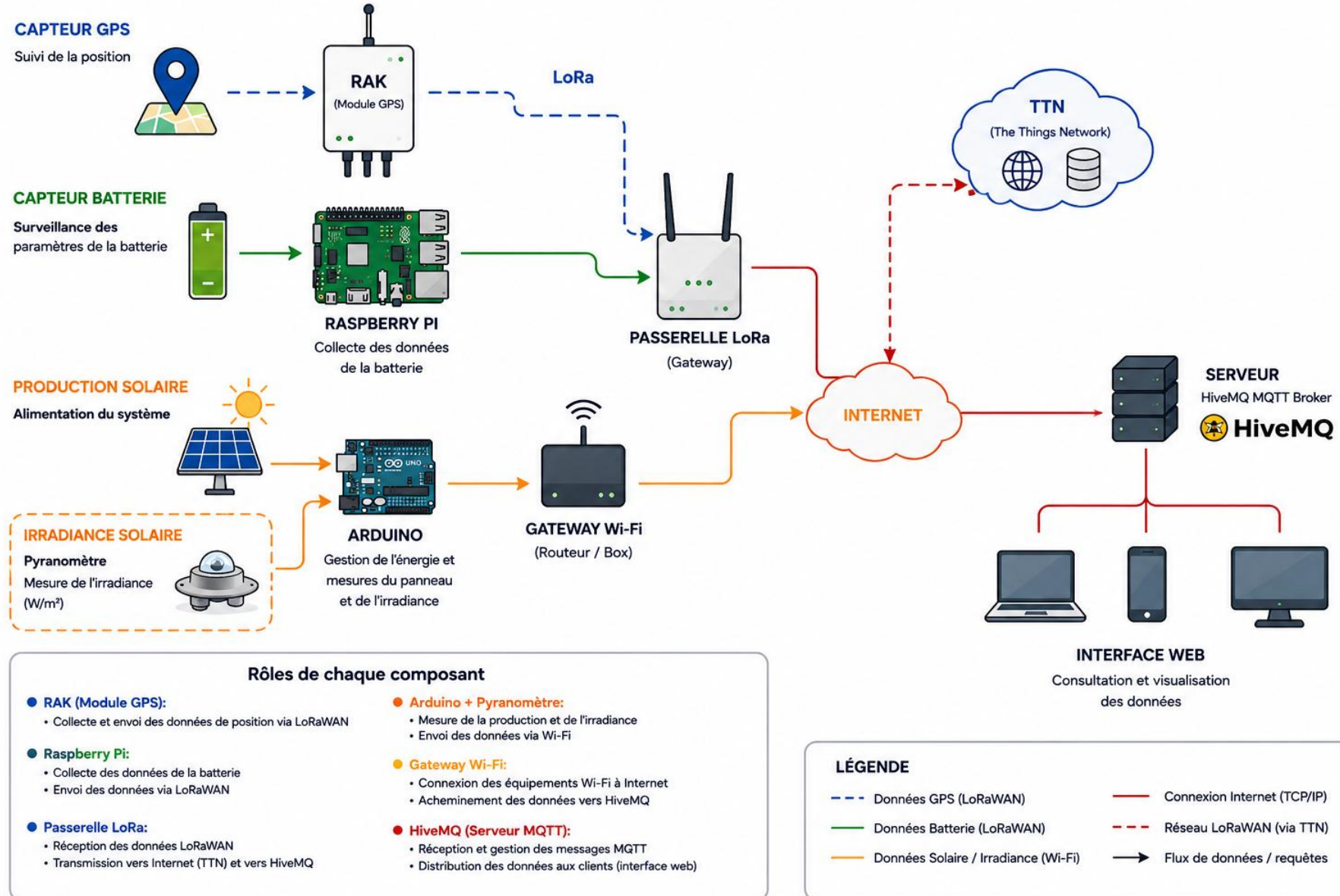


Diagramme de Cas d'utilisation (perso)

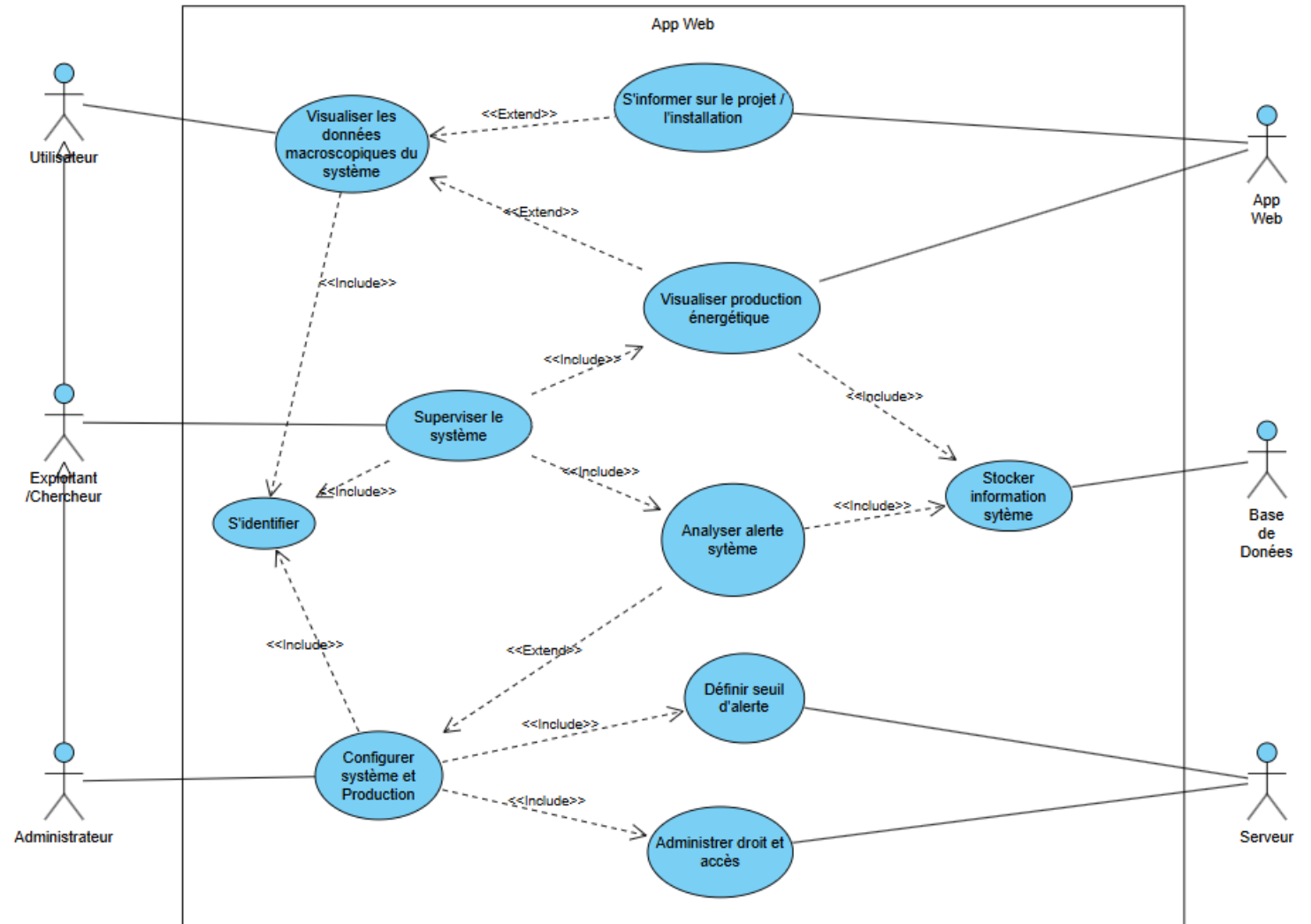
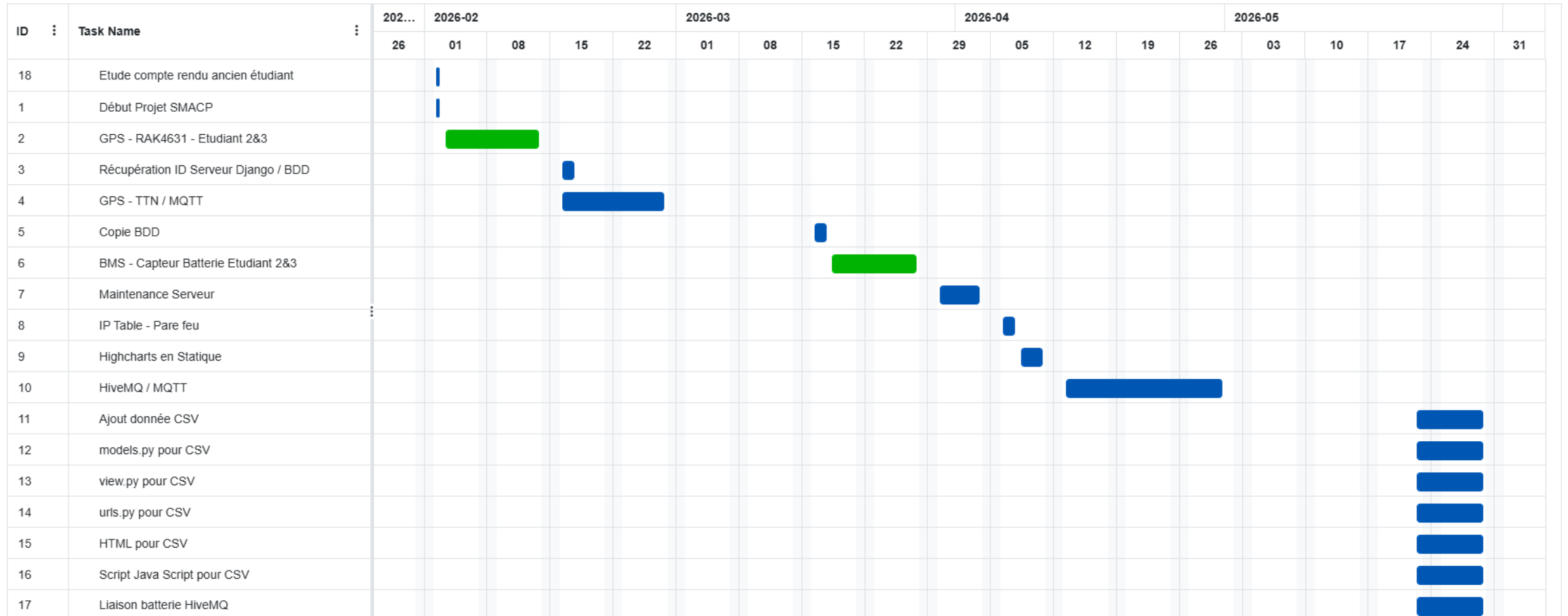


Diagramme de Gantt

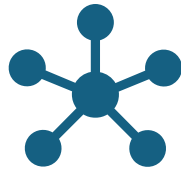


Powered by: onlinegantt.com

Objectifs du Projet (partie personnelle)



Intégrer les nouvelles fonctionnalités de mes collègues 2026



Intégrer toutes les données N0 à N7 (actuellement seulement N6)

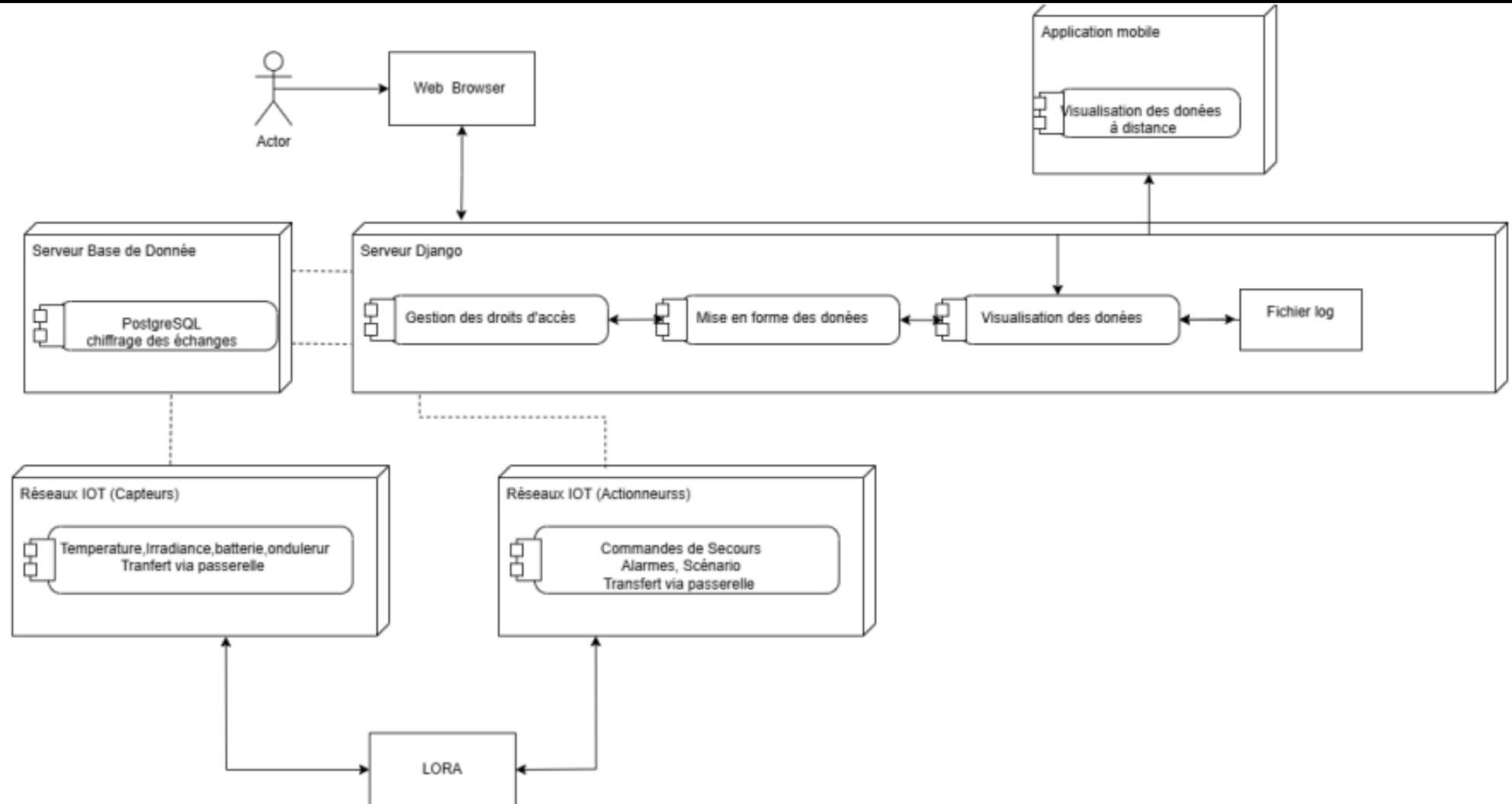


Vérification des règles de firewall



Reprise du GitHub à rendre « conforme » à un versionning

Architecture Générale



Technologies utilisées

- Python
- Django
- PostgreSQL
- MQTT
- HiveMQ
- Ubuntu 24.10
- Nginx
- Gunicorn
- Vs Code



Pourquoi Django ?



Rapide a mettre en place



Très complet



Sécurisé



Evolutif et Polyvalent

Fonctionnement de l'application

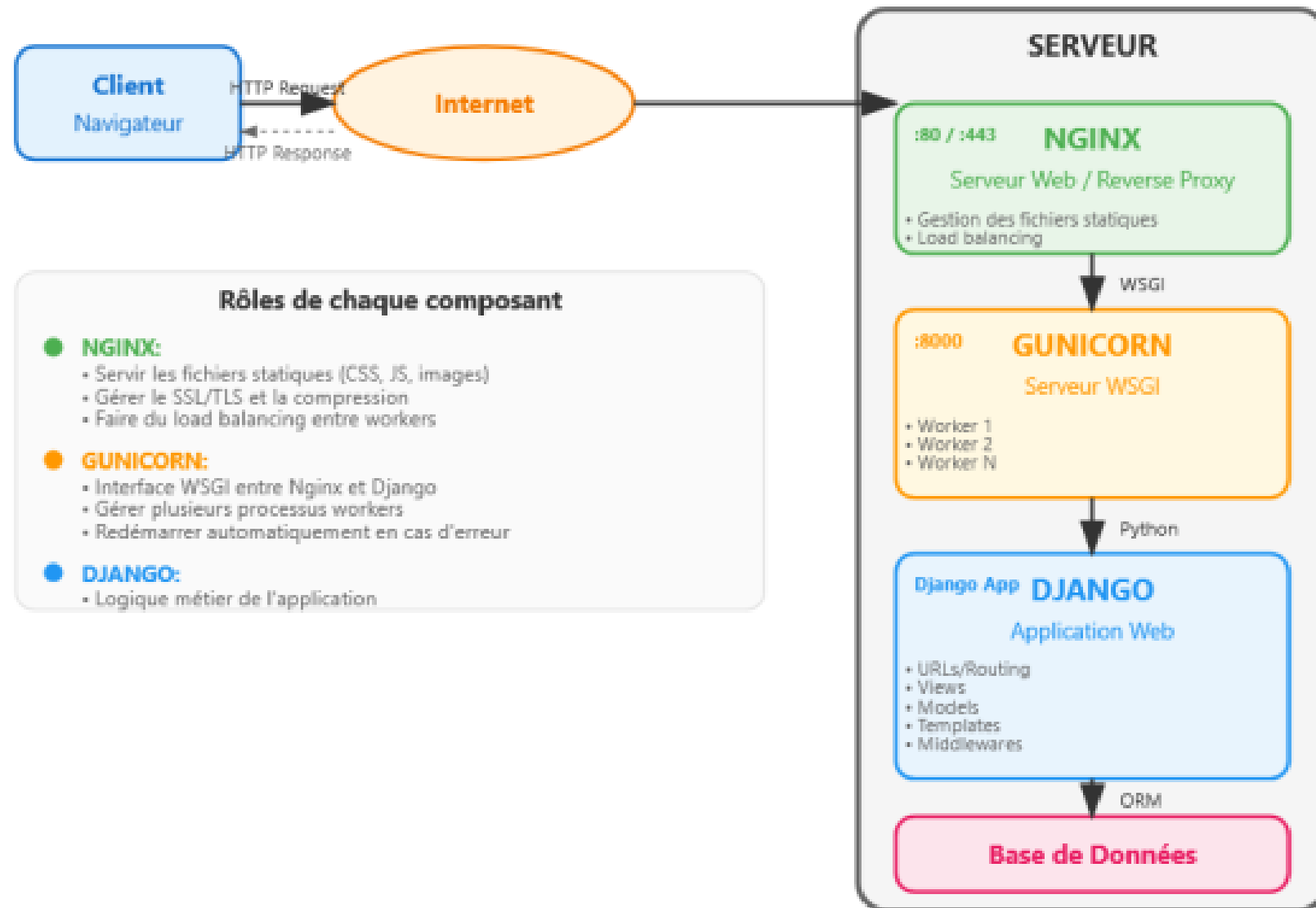
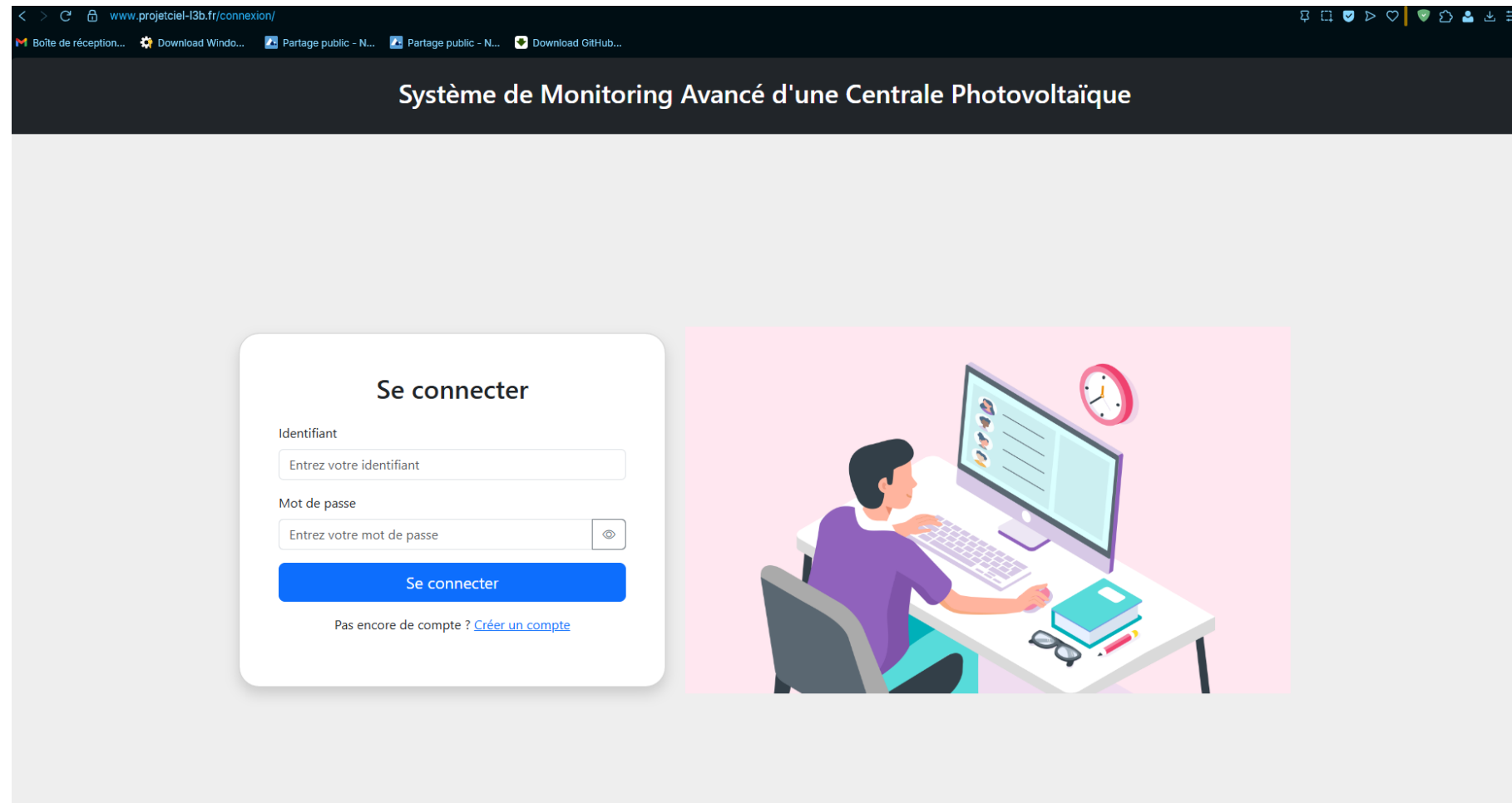


Schéma structurel de la communication

Interface Utilisateur



Page d'accueil du site

🏠 Accueil

📅 Tableau des données

Graphiques

📊 Nzero

📊 Nun

📊 Ndeux

📊 Ntrois

📊 Nquatre

📊 Ncinq

📊 Nsix

Période :

-- Choisir une période --

jj/mm/aaaa --:--



jj/mm/aaaa --:--

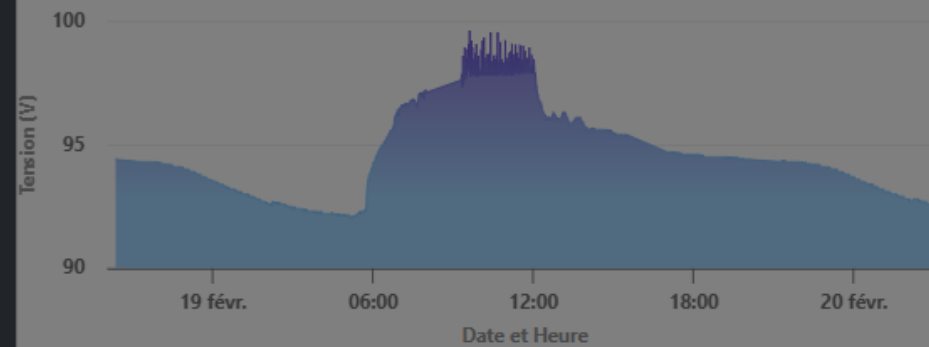


Filtrer

☒ Tension

Tension (V)

Cliquez-glissez pour zoomer

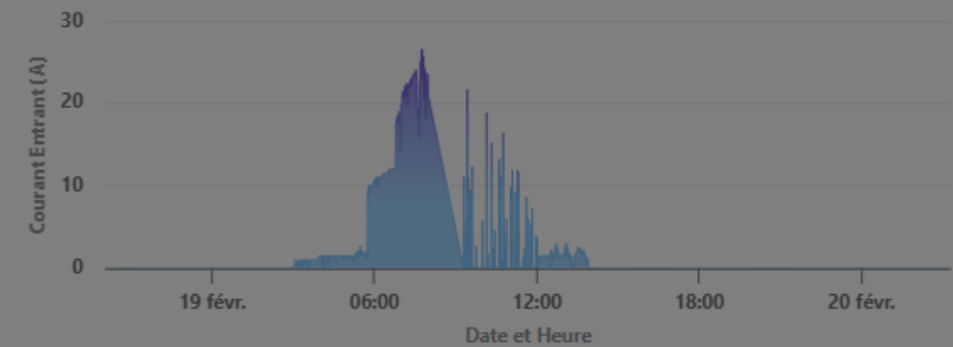


Highcharts.com

☒ Courant Entrant

Courant Entrant (A)

Cliquez-glissez pour zoomer

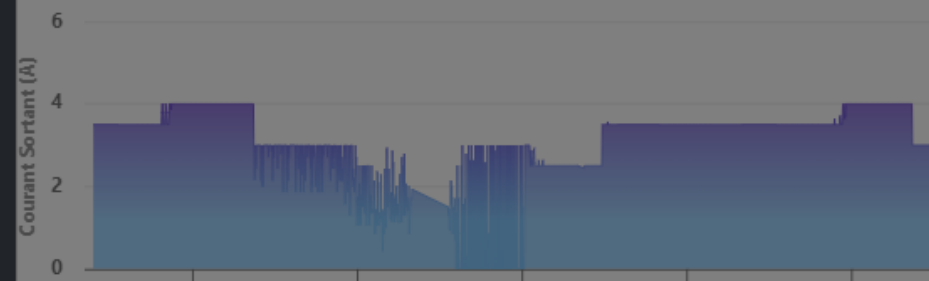


Highcharts.com

☒ Courant Sortant

Courant Sortant (A)

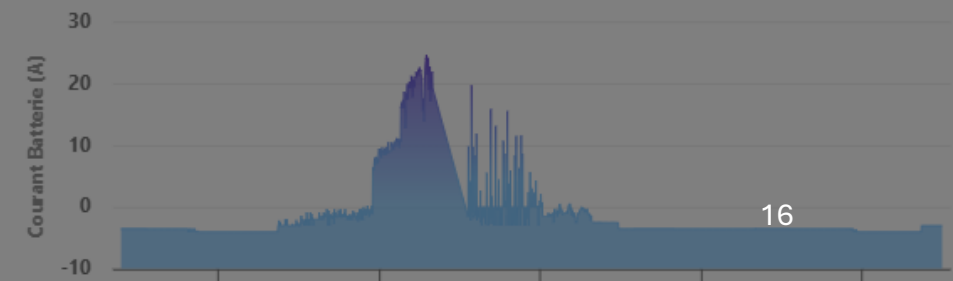
Cliquez-glissez pour zoomer



☒ Courant Batterie

Courant Batterie (A)

Cliquez-glissez pour zoomer



Difficultés rencontrées / Correctifs & Travail réaliser

- Documentation incomplète / peu détaillée
- Git push défectueux
- Liens Highcharts invalides
- Gunicorn
- Intégration donnée Batterie → Django
- Liason Batterie → HiveMQ → Django

```
ubuntu@vps-62f78ac3:~$ sudo systemctl status gunicorn
* gunicorn.service - gunicorn daemon
   Loaded: loaded (/etc/systemd/system/gunicorn.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Wed 2026-04-01 19:32:32 UTC; 14h ago
     Duration: 135ms
  Invocation: 18a714d499f54382bcf817927be0a491
 TriggeredBy: * gunicorn.socket
    Process: 1385 ExecStart=/home/ubuntu/dev/envdjango/bin/gunicorn --access-logfile - --workers 3 --bind unix:/run/gunicorn.sock monProjetDjango.wsgi:application (code=exited, status=3)
   Main PID: 1385 (code=exited, status=3)

Apr 01 19:32:31 vps-62f78ac3 gunicorn[1386]: ModuleNotFoundError: No module named 'monProjetDjango.wsgi'
Apr 01 19:32:31 vps-62f78ac3 gunicorn[1386]: [2026-04-01 19:32:31 +0000] [1386] [INFO] Worker exiting (pid: 1386)
Apr 01 19:32:31 vps-62f78ac3 gunicorn[1385]: [2026-04-01 19:32:31 +0000] [1385] [ERROR] Worker (pid:1386) exited with code 3
Apr 01 19:32:31 vps-62f78ac3 gunicorn[1385]: [2026-04-01 19:32:31 +0000] [1385] [ERROR] Shutting down: Master
Apr 01 19:32:31 vps-62f78ac3 gunicorn[1385]: [2026-04-01 19:32:31 +0000] [1385] [ERROR] Reason: Worker failed to boot.
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: gunicorn.service: Main process exited, code=exited, status=3/NOTIMPLEMENTED
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: gunicorn.service: Failed with result 'exit-code'.
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: gunicorn.service: Start request repeated too quickly.
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: gunicorn.service: Failed with result 'exit-code'.
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: Failed to start gunicorn.service - gunicorn daemon.
ubuntu@vps-62f78ac3:~$
```

```

8     <meta charset="UTF-8">
9     <title>Dashboard</title>
10    <link rel="stylesheet" href="{% static 'css/styles.css' %}">
11    <script src="https://code.highcharts.com/highcharts.js"></script>
12    <script src="https://code.highcharts.com/modules/exporting.js"></script>
13    <script src="https://code.highcharts.com/modules/export-data.js"></script>
14    <script src="https://code.highcharts.com/modules/accessibility.js"></script>
15    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
16 </head>

```

Liens CDN Highcharts

```

<head>
    <meta charset="UTF-8">
    <title>Dashboard</title>
    <link rel="stylesheet" href="{% static 'css/styles.css' %}">
    <script src="{% static 'js/highcharts.js' %}"></script>
    <script src="{% static 'js/exporting.js' %}"></script>
    <script src="{% static 'js/export-data.js' %}"></script>
    <script src="{% static 'js/accessibility.js' %}"></script>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

```

Liens statique Highcharts

Graphique des données

Menu

Période :

-- Choisir une période --

jj/mm/aaaa --:--

jj/mm/aaaa --:--

Filtrer

Choisir les paramètres :

Courant Batterie (A)

The screenshot shows the Chrome DevTools Network tab. The top panel is a timeline of requests. The bottom panel shows the details for a request to 'https://code.highcharts.com/12.5.0/highcharts.js'. The request is a GET method with a status of 403 Forbidden. The response headers show 'Cache-Control: private, max-age=0, no-store, no-cache, must-revalidate, post-check=0, pre-check=0' and 'Content-Type: text/html; charset=UTF-8'. The console shows a warning message: 'Warning! This area is for use by developers only. Scammers have been known to encourage people to copy/paste information here to hack accounts. Do not proceed if you are unsure.'

Name	Request URL	Request Method	Status Code	Referrer Policy
dashboard/				
isolated-first.js				
bootstrap.min.css				
bootstrap.bundle.min.js				
styles.css				
highcharts.js	https://code.highcharts.com/12.5.0/highcharts.js	GET	403 Forbidden	same-origin
exporting.js				
export-data.js				
accessibility.js				
bootstrap.min.css				
data:image/svg+xml,...				
data:image/svg+xml,...				
favicon.ico				

Cache-Control	Cf-Ray	Content-Encoding	Content-Type	Date	Expires	Referrer-Policy	Server	Vary	X-Frame-Options
private, max-age=0, no-store, no-cache, must-revalidate, post-check=0, pre-check=0	9ecac807ef24f01f-JNB	gzip	text/html; charset=UTF-8	Wed, 15 Apr 2026 11:58:57 GMT	Thu, 01 Jan 1970 00:00:01 GMT	same-origin	cloudflare	accept-encoding	SAMEORIGIN

:authority	:method	:path	:scheme
code.highcharts.com	GET	/12.5.0/highcharts.js	http

13 requests | 17.0 kB transferred

Warning! This area is for use by developers only. Scammers have been known to encourage people to copy/paste information here to hack accounts. Do not proceed if you are unsure.

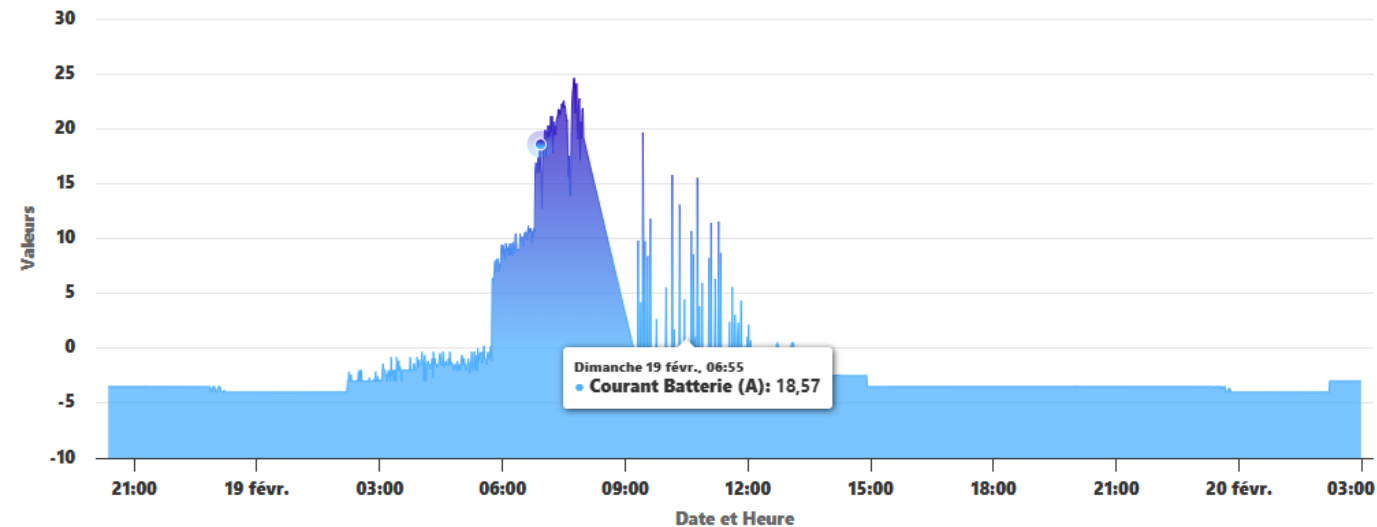
GET https://www.projetciel-13b.fr/favicon.ico 404 (Not Found) favicon.ico:1

Graphique des données

Menu

Évolution des Mesures

Cliquez-glissez pour zoomer



Courant Batterie (A)

Période :

-- Choisir une période --

jj/mm/aaaa --:--

jj/mm/aaaa --:--

Filtrer

Choisir les paramètres :

Courant Batterie (A)

Elements Console Sources Network Performance Memory Application

Filter

All Fetch/XHR Doc CSS JS Font Img Media Manifest Socket Wasm Other

Name

dashboard/

bootstrap.bundle.min.js

styles.css

highcharts.js

exporting.js

export-data.js

accessibility.js

bootstrap.min.css

bootstrap.min.css

isolated-firstjst

data:image/svg+xml,...

data:image/svg+xml,...

General

Request URL https://www.projetciel-l3b.fr/static/js/highcharts.js

Request Method GET

Status Code 200 OK (from memory cache)

Remote Address [64:ff9b::335b:f98b]:443

Referrer Policy same-origin

Response headers

Accept-Ranges bytes

Connection keep-alive

Content-Length 275796

Content-Type application/javascript

Date Thu, 16 Apr 2026 07:52:40 GMT

Etag "69df8328-43554"

Last-Modified Wed, 15 Apr 2026 12:23:04 GMT

Server nginx/1.26.0 (Ubuntu)

Request Headers

Referer https://www.projetciel-l3b.fr/espace_chercheur/dashboard/

Sec-Ch-Ua "Not:A-Brand",v="99", "Opera GX",v="129", "Chromium",v="145"

Sec-Ch-Ua-Mobile ?0

12 requests 16.6 kB transferred

Console What's new

Warning!

This area is for use by developers only. Scammers have been known to encourage people to copy/paste information here to hack accounts. Do not proceed if you are unsure.

Uncaught SyntaxError: Unexpected token 'export' (at webpage_content_reporter.js:1:115561 webpage_content_reporter.js:1:115561)

Integration données Batterie → Django

Address:	Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
N+0	Total voltage		Current IN		Current OUT		Current Battery	
N+1	Energy stored				Battery capacity		SOC	-
N+2	Energy today collected				Energy today consumed			
N+3	Total energy collected				Total energy consumed			
N+4	Cell voltage MIN		Cell voltage MAX		Cell voltage Bypass		-	-
N+5	Cell voltage Lowest		Low Nr	Cell voltage Highest		High Nr	Sbyte 1	Sbyte2
N+6	Tmp,Lowest	Low Nr	Tmp,Highest	High Nr	Min charg temp	Min dis temp	Max temp	-
N+7	Current cell voltage		Cur.Temp	Cur. Nr	Cell cnt	-	Isolation resistance	

Donnée CSV → Django

- Models.py
- View.py
- Urls.py
- Fichier HTML
- Fichier JavaScript

```
class Donnee(models.Model):
    date = models.DateTimeField(primary_key=True)
    total_volt = models.FloatField()
    currentin = models.FloatField()
    currentout = models.FloatField()
    current_bat = models.FloatField()

    class Meta:
        managed = False
        db_table = 'current'

class Nzero(models.Model):
    date = models.DateTimeField(primary_key = True)
    total_volt = models.FloatField()
    currentin = models.FloatField()
    currentout = models.FloatField()
    current_bat = models.FloatField()
    class Meta:
        managed = False
        db_table = 'DB_nzero'
```

Models.py

```
#ajout 2026
path('grapheDesDonnees_nun/', views.GrapheDesDonnees_nun, name='visualisation_graphique_nun'),
path('grapheDesDonnees_ndeux/', views.GrapheDesDonnees_ndeux, name='visualisation_graphique_ndeux'),
path('grapheDesDonnees_ntrois/', views.GrapheDesDonnees_ntrois, name='visualisation_graphique_ntrois'),
path('grapheDesDonnees_nquatre/', views.GrapheDesDonnees_nquatre, name='visualisation_graphique_nquatre'),
path('grapheDesDonnees_ncinq/', views.GrapheDesDonnees_ncinq, name='visualisation_graphique_ncinq'),
path('grapheDesDonnees_nsix/', views.GrapheDesDonnees_nsix, name='visualisation_graphique_nsix'),
]
```

Urls.py


```

43 def GrapheDesDonnees(request):
44     data = list(Donnee.objects.values('date', 'total_volt', 'currentin', 'currentout', 'current_bat'))
45     data_json = json.dumps(data, default=str) # Convertir QuerySet en JSON
46     return render(request, "visualisation_graphique.html", {'data_json': data_json})
47
48 def GrapheDesDonnees_nun(request):
49     data = list(Nun.objects.values('date', 'Energy_stored_Wh', 'Battery_capacity_kWh', 'SOC_percent', 'nn'))
50     data_json = json.dumps(data, default=str)
51     return render(request, "visualisation_graphique_nun.html", {'data_json': data_json})
52
53 def GrapheDesDonnees_ndeux(request):
54     data = list(Ndeux.objects.values('date', 'Energy_today_collected_kWh', 'Energy_today_consummed_kWh'))
55     data_json = json.dumps(data, default=str) # Convertir QuerySet en JSON
56     return render(request, "visualisation_graphique_ndeux.html", {'data_json': data_json})
57
58 def GrapheDesDonnees_ntrois(request):
59     data = list(Ntrois.objects.values('date', 'Total_Energy_today_collected_kWh', 'Total_Energy_today_consummed_kWh'))
60     data_json = json.dumps(data, default=str)
61     return render(request, "visualisation_graphique_ntrois.html", {'data_json': data_json})
62
63 def GrapheDesDonnees_nquatre(request):
64     data = list(Nquatre.objects.values('date', 'Cell_voltage_min_V', 'Cell_voltage_max_V', 'Cell_voltage_bypass_V', 'nonnommeeun', 'nonnommeedeux', 'nonnommeetrois'))
65     data_json = json.dumps(data, default=str)
66     return render(request, "visualisation_graphique_nquatre.html", {'data_json': data_json})
67
68 def GrapheDesDonnees_ncinq(request):
69     data = list(Ncinq.objects.values('date', 'Cell_voltage_lowest_V', 'Low_nr', 'Cell_voltage_highest_V', 'High_nr', 'Sbyteun', 'Sbytedeux'))
70     data_json = json.dumps(data, default=str)
71     return render(request, "visualisation_graphique_ncinq.html", {'data_json': data_json})
72
73 def GrapheDesDonnees_nsix(request):
74     data = list(Nsix.objects.values('date', 'Tmp_lowest_deg', 'Tmp_highest_deg', 'Min_charg_temp_deg', 'Max_temp_deg'))
75     data_json = json.dumps(data, default=str) # Convertir QuerySet en JSON
76     return render(request, "visualisation_graphique_nsix.html", {'data_json': data_json})

```

Views.py

Donnée Batterie sur HiveMQ

🟢 The WebClient is connected

Connection Settings

Connect to your HiveMQ Cloud Cluster with your credentials. Or, connect instantly with autogenerated credentials.

Username

bms_user

Password

Monpassword123

🔒

Disconnect

Topic Subscriptions

1

Unsubscribe from all topics

Subscribe to topics to receive messages from the HiveMQ cluster. You can also set the Quality of Service (QoS) for each topic. The higher the QoS, the more reliable the message delivery is. You can always subscribe to the  wildcard to receive all messages. Please note that the messages from internal probe topics are not displayed here.

Topic	QoS	Actions
📡 #	QoS: 0	🗑️

Messages

64

🔍

↶↷

Clear all messages

0

Topic: bms/N0

QoS: 0

{"total_voltage_V": 25.5, "current_in_A": 0.0, "current_out_A": 0.0, "current_bat_A": 0.0, "timestamp": "2026-06-01T16:13:18.979415"}

1

Topic: bms/N2

QoS: 0

{"Energy_today_collected_kWh": 0, "Energy_today_consummed_kWh": 0, "timestamp": "2026-06-01T16:13:18.981882"}

2

Topic: bms/N4

QoS: 0

{"Cell_voltage_min_V": 3.0, "Cell_voltage_max_V": 3.8, "Cell_voltage_bypass_V": 0.013, "unused1": 72.0, "unused2": -20.0, "unused3": 75.0, "timestamp": "2026-06-01T16:13:18.985340"}

3

Topic: bms/N6

QoS: 0

{"Tmp_lowest_deg": 17, "Low_nr": 8, "Tmp_highest_deg": 24, "High_nr": 6, "Min_charg_temp_deg": 0, "Min_dis_temp_deg": -20, "Max_temp_deg": 75, "unused": 0, "timestamp": "2026-06-01T16:13:18.989491"}

4

Topic: bms/N1

QoS: 0

{"Energy_stored_Wh": 496, "Battery_capacity_kWh": 13, "SOC_percent": 38, "unused": 0, "timestamp": "2026-06-01T16:13:18.980829"}

5

Topic: bms/N5

QoS: 0

{"Cell_voltage_lowest_V": 2.645, "Low_nr": 1.0, "Cell_voltage_highest_V": 3.325, "High_nr": 7.0, "Sbyte1": -120.0, "Sbyte2": 0.0, "timestamp": "2026-06-01T16:13:18.987611"}

6

Topic: bms/cell/C5

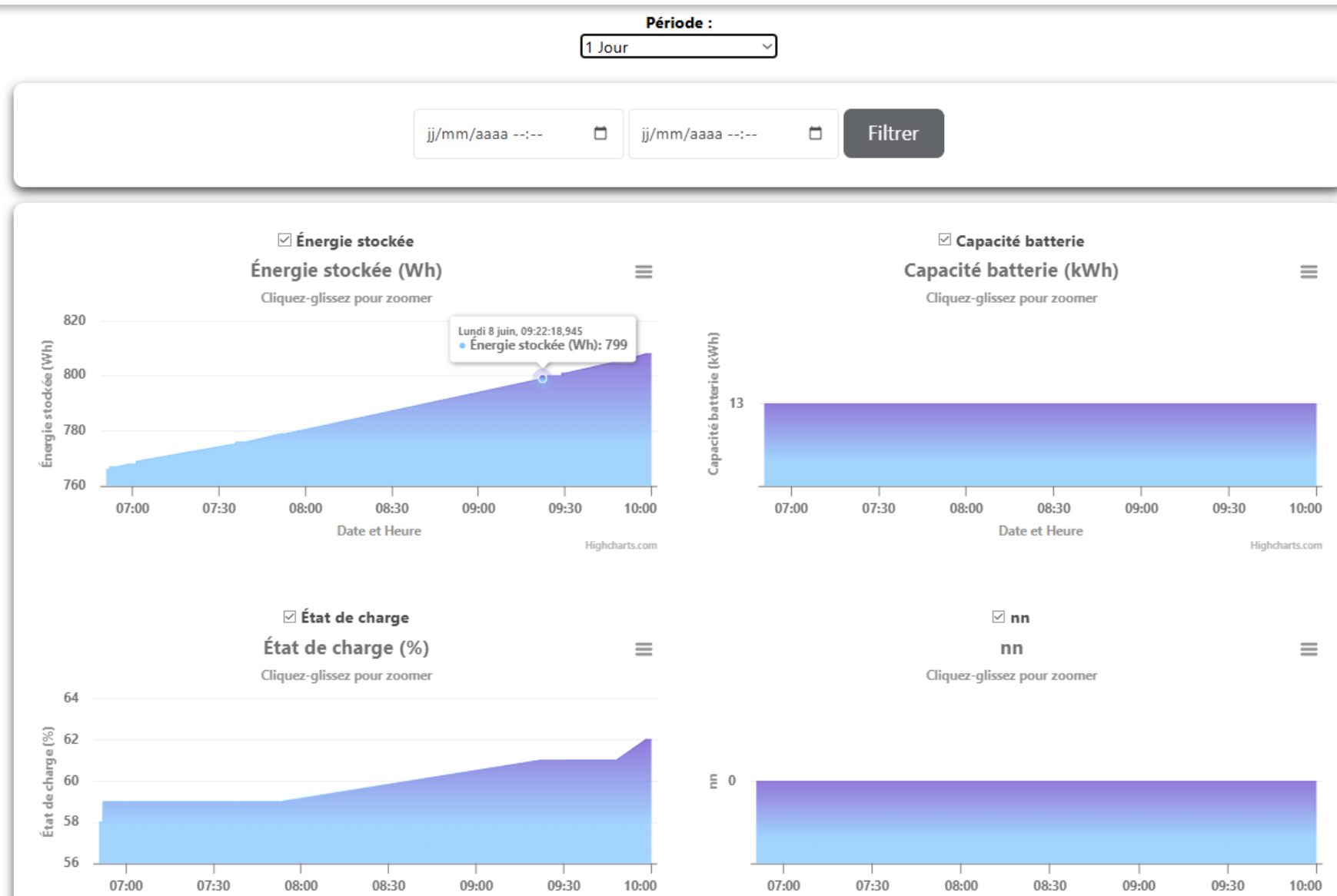
QoS: 0

{"Current_Cell_voltage_V": 3.265, "Curr_temp_deg": 22.0, "Curr_nr": 5.0, "Cell_cnt": 8.0, "unused1": 0.0, "Isolation_resistance_kOhm": 3.0, "unused2": -25.0, "timestamp": "2026-06-01T16:13:18.992227"}

Données récupérées sur Django

```
[2026-06-08 14:00:09,407: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[5884d228-f647-4929-adf0-9974e785cf23] received
[2026-06-08 14:00:09,422: INFO/ForkPoolWorker-4] BMS enregistré – topic: bms/N6 | ts: 2026-06-08 14:00:00.542380+04:00
[2026-06-08 14:00:09,423: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[5884d228-f647-4929-adf0-9974e785cf23] succeeded in 0.016127551905810833s: None
[2026-06-08 14:00:09,931: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[edb2122a-aa50-4310-84d3-26a24a64c490] received
[2026-06-08 14:00:09,948: INFO/ForkPoolWorker-4] BMS enregistré – topic: bms/N0 | ts: 2026-06-08 14:00:01.511685+04:00
[2026-06-08 14:00:09,948: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[edb2122a-aa50-4310-84d3-26a24a64c490] succeeded in 0.016321729868650436s: None
[2026-06-08 14:00:10,213: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[7b254712-d29c-49db-bd1f-34159eba05a9] received
[2026-06-08 14:00:10,230: INFO/ForkPoolWorker-4] BMS enregistré – topic: bms/N2 | ts: 2026-06-08 14:00:01.514042+04:00
[2026-06-08 14:00:10,230: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[7b254712-d29c-49db-bd1f-34159eba05a9] succeeded in 0.0162708000279963s: None
[2026-06-08 14:00:10,244: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[e7bfdac6-6ace-4d06-947d-86529735e139] received
[2026-06-08 14:00:10,260: INFO/ForkPoolWorker-4] BMS enregistré – topic: bms/N6 | ts: 2026-06-08 14:00:01.520412+04:00
[2026-06-08 14:00:10,261: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[e7bfdac6-6ace-4d06-947d-86529735e139] succeeded in 0.015957661904394627s: None
[2026-06-08 14:00:10,271: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[6e0f76b7-784a-4810-b5c2-ae9e88f3c3b6] received
[2026-06-08 14:00:10,286: INFO/ForkPoolWorker-4] BMS enregistré – topic: bms/N4 | ts: 2026-06-08 14:00:01.517316+04:00
[2026-06-08 14:00:10,287: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[6e0f76b7-784a-4810-b5c2-ae9e88f3c3b6] succeeded in 0.015664461068809032s: None
[2026-06-08 14:00:10,292: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[a45b75d8-01a5-4864-b640-d6a28527433a] received
[2026-06-08 14:00:10,308: INFO/ForkPoolWorker-4] BMS enregistré – topic: bms/N1 | ts: 2026-06-08 14:00:01.513023+04:00
[2026-06-08 14:00:10,309: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[a45b75d8-01a5-4864-b640-d6a28527433a] succeeded in 0.01633399398997426s: None
[2026-06-08 14:00:10,324: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[cb96559b-0059-47b3-bf4d-b141c9a2fcc5] received
[2026-06-08 14:00:10,340: INFO/ForkPoolWorker-4] BMS enregistré – topic: bms/N3 | ts: 2026-06-08 14:00:01.514998+04:00
[2026-06-08 14:00:10,340: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[cb96559b-0059-47b3-bf4d-b141c9a2fcc5] succeeded in 0.01599380187690258s: None
[2026-06-08 14:00:10,366: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[c9f5d2a6-0395-4ee6-8b0f-16e5f0294287] received
[2026-06-08 14:00:10,382: INFO/ForkPoolWorker-4] BMS enregistré – topic: bms/N5 | ts: 2026-06-08 14:00:01.519155+04:00
[2026-06-08 14:00:10,382: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[c9f5d2a6-0395-4ee6-8b0f-16e5f0294287] succeeded in 0.015906693879514933s: None
```

Donnée Batterie sur le site



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○

Sécurisation du serveur Django

iptables / UFW

- Filtrage des connexions réseau.
- Autorisation uniquement des services nécessaires.
- Blocage des ports inutilisés.

Fail2ban

- Surveillance des tentatives de connexion.
- Détection des échecs répétés d'authentification.
- Bannissement automatique des adresses IP suspectes.

Bénéfices

- Réduction de la surface d'attaque.
- Protection contre les attaques par force brute.
- Meilleur contrôle des accès réseau.

Conclusion



Objectifs atteints



Optimisation
restante



Compétence
acquise



Suite du projet